

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591 -> 2.380505241	delta=+0.0289076
ratio=1.0122928		
CoG x [m]	-0.002024708562 -> -0.002154223971	delta=-0.000129515
ratio=-		
CoG y [m]	-3.052657894e-05 -> 0.001119685915	delta=+0.00115021
ratio=-		
CoG z [m]	0.009509749599 -> 0.01093715174	delta=+0.0014274
ratio=-		
rotor effectiveness 1	1 -> 0.8961259081	delta=-0.103874 ratio=0.89612591
rotor effectiveness 2	1 -> 0.9121652073	delta=-0.0878348 ratio=0.91216521
rotor effectiveness 3	1 -> 0.8841798525	delta=-0.11582 ratio=0.88417985
rotor effectiveness 4	1 -> 0.8865342506	delta=-0.113466 ratio=0.88653425
command lag [s]	0 -> 0.03866229884	delta=+0.0386623 ratio=-

Inertia matrix [kg m^2]

reference:

[0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[1.901508003e-05, 5.9167305e-08, 0.1289921107]

estimated:

[0.06496532087, -4.89455007e-05, -2.042475642e-05]
[-4.89455007e-05, 0.06525753549, 0.0001065039808]
[-2.042475642e-05, 0.0001065039808, 0.1289515182]

Principal moments: reference -> estimated

principal moment 1 [kg m^2]	0.06495264516 -> 0.06495734037	delta=+4.69521e-06
ratio=1.0000723		
principal moment 2 [kg m^2]	0.06500006701 -> 0.06526533133	delta=+0.000265264
ratio=1.004081		
principal moment 3 [kg m^2]	0.1289921163 -> 0.1289517029	delta=-4.04134e-05
ratio=0.9996867		

Optimization

joint residual-wrench accumulated squared loss: 4.79872320407
Gaussian-prior cost: 2.42591501275
selected strict-ZOH lag [s]: 0.038662298837

Bag failure_1_single

reconstruction position RMSE [m]: 0.00049721264017
reconstruction position component RMSE [m]: [0.000302 0.000297 0.000261]
reconstruction orientation RMSE [deg]: 0.528821719301
gyro consistency RMSE xyz [rad/s]: [0.060839 0.087229 0.04255]
specific-force consistency RMSE xyz [m/s^2]: [0.933205 0.203336 0.378942]
inferred external force RMS xyz [N]: [0.715939 0.496653 1.537207]
inferred external torque RMS xyz [N m]: [0.109217 0.445336 0.207172]
legacy acceleration-residual loss diagnostic: 0.828409375077

Recorded command timestamp intervals (data-derived)

Bag failure_1_single

rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s
gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s

Parameter data objective

sample-summed uncentered squared dimensionless residual body wrench
force scale F^* = 23.0612945139 N
torque scale τ^* = 4.41819797713 N m
residual mean is NOT subtracted before optimization

Selected optimizer diagnostics

status=1 success=True nfev=4 njev=4
message='gtol' termination condition is satisfied.
reported optimality=6.174483937471109e-07
initial cost=7.224679132875364
final cost=7.224638216820725
cost reduction=4.0916054639517085e-05
coordinate step L2=0.011148276663852536
final gradient inf-norm=6.174483937471109e-07
gradient tolerance satisfied=True
final Jacobian rank=14
final Jacobian cond(nonzero)=590.5038145477506
max directional finite-difference Jacobian relative error=6.002192621980195e-12